

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (CE)

Dec 2012

CE711 Information & Network Security

Date: 11.12.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Write the difference between poly alphabetic and mono alphabetic ciphers. [05]
(b) Explain the one time pad cipher in a brief. [02]
- Q - 2 (a) Define the three security goals. [04]
(b) Explain the working of play fair cipher with example. [07]
(c) Compare substitution cipher with transposition cipher. [03]

OR

- Q - 2 (a) Explain the following active attacks: [04]
a. Masquerade
b. Replay
c. Modification of message
d. Denial of service
- (b) Explain working of Hill cipher with example. [07]
(c) What is chosen plain text attack? Explain with diagram. [03]
- Q - 3 (a) What is DES? Explain working of it. [06]
(b) Explain Counter mode block cipher operation. [04]
(c) Explain any one component of modern block cipher. [04]

OR

- Q - 3 (a) Why AES is stronger cipher compare to DES? Explain encryption steps of AEC cipher. [06]
(b) What is block cipher operations? [04]
(c) What is stream cipher? Give example of it. [04]

SECTION – II

Q - 4 (a) Differentiate steganography and cryptography. [03]

(b) What is Secure Socket Layer (SSL)? Is SSL a part of Transport Layer Security? [04]

Justify your answer.

Q - 5 (a) List and explain the applications of cryptographic hash functions. [07]

(b) Explain the working of RSA algorithm. Why only prime numbers are used in RSA. [07]

Justify your answer.

OR

(b) What is key distribution? Explain Diffie-Hellman key exchange algorithm with [07]

suitable example.

Q - 6 Write a short note on any TWO. [14]

a. X.509

b. Message Authentication Code

c. SMIME
